Chapter 11

Algebraic Ratio and Proportion

It is very important for us to have a clear conception of ratio and proportion in mathematics. Arithmetical ratio and proportion have been elaborately discussed in class VII. In this chapter, algebraic ratio and proportion will be discussed. We regularly use the concept of ratio and proportion in many applications: the production of construction materials, food staff and goods , applying fertilizer in land, making beautiful shapes and design to make things attractive and good looking, and so on. Many problems of daily lives can also be solved by using the concept of ratio and proportion.

After studying this chapter, the students will be able to —

- explain algebraic ratio and proportion.
- use different types of rules of transformation related to proportion.
- describe successive proportion.
- use ratio, proportion, successive proportion in solving real life problems.

Ratio and Proportion

Ratio

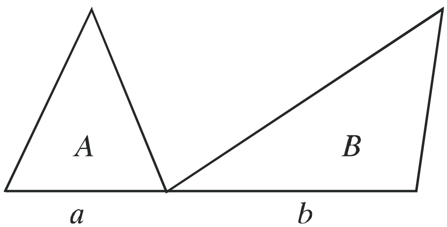
If we have two quantities (or numbers) of the same unit, we can express one quantity as a multiple (or a fraction) of the other quantity. This fraction is called the ratio of two quantities.

The ratio of two quantities p and q is written as $p : q = \frac{p}{q}$. These two quantities, p and q are of same kind and same unit. Here, p is called antecedent and q is called subsequent of the ratio.

Sometimes we use the ratio as an approximate measure. For example, the number of cars on the road at 10 A.M. is two times higher than the number of cars at 8 A.M. In this case, it is not necessary to know the exact number of cars to determine the ratio. Again, in many occasions, we say that the area of your house is three times to the area of mine. Also in this case, it is not necessary to know the exact area of the house. We use the concept of ratio in many other cases in real life.

Proportion

If four quantities (or numbers), are such that the ratio of the first and the second quantities is equal to the ratio of the third and the fourth quantities, those four quantities form a proportion. If a, b, c, d are four such quantities, we write $a : b = c : d$. The four quantities need not to be of same kind. The two quantities of the ratio need to be of the same kind.



In the above figure, let the bases of two triangles be a and b, respectively and each of their height is h unit. If the areas of these triangle are A and B square units, respectively, we can write:

$$\frac{A}{B} = \frac{\frac{1}{2}ah}{\frac{1}{2}bh} = \frac{a}{b} \qquad \text{or, } A : B = a : b$$

i.e. the ratio of two areas is equal to the ratio of two bases.

Continued proportion

By continued proportion of a, b, c, it is meant that a : b = b : c. Here a, b, c will be continued proportion, if and only if $b^2 = ac$. In case of continued proportional, all the quantities are to be of same kind. In this case, c is called the third proportional of a and b, and b is called the mid-proportional of a and c.

2026

Example 1. A and B traverse the fixed distance in t_1 and t_2 minutes. Find the ratio of the average velocity of A and B

Solution: Let the average velocity of A and B be v_1 metre and v_2 metre per minute respectively. So, in time t_1 minutes A traverses v_1t_1 metre and in t_2 minutes B traverses v_2t_2 metra. According to the problem, $v_1t_1 = v_2t_2$ $\therefore \frac{v_1}{v_2} = \frac{t_2}{t_1}$ Here, the ratio of the velocities is inversely proportional to the ratio of time.

Work:

1) Express 3.5 : 5.6 into 1 : a and b : 1.
2) If $x : y = 5 : 6$, then the value of $3x : 5y = ?$

Transformation of ratio

Here, the quantities of ratios are positive numbers.

1. if $a : b = c : d$, then $b : a = d : c$ [Invertendo]

   Proof: Given that,
   $\frac{a}{b} = \frac{c}{d}$
   or, $ad = bc$ [multiplying both the sides by bd]
   or, $\frac{ad}{ac} = \frac{bc}{ac}$ [dividing both the sides by ac where a and c are non-zero]
   or, $\frac{d}{c} = \frac{b}{a}$
   i.e, $b : a = d : c$

2. If $a : b = c : d$, then, $a : c = b : d$ [Alternendo]

   Proof: Given that,
   $\frac{a}{b} = \frac{c}{d}$
   or, $ad = bc$ [multiplying both the sides by bd]
   or, $\frac{ad}{cd} = \frac{bc}{cd}$ [dividing both the sides by cd where c, d are non-zero]
   or, $\frac{a}{c} = \frac{b}{d}$
   i.e., $a : c = b : d$

2026

3. if $a : b = c : d$, then, $\dfrac{a+b}{b} = \dfrac{c+d}{d}$ [Componendo]

Proof: Given that,

$\dfrac{a}{b} = \dfrac{c}{d}$

or, $\dfrac{a}{b} + 1 = \dfrac{c}{d} + 1$ [adding 1 to both the sides]

i.e, $\dfrac{a+b}{b} = \dfrac{c+d}{d}$

4. If $a : b = c : d$, then $\dfrac{a-b}{b} = \dfrac{c-d}{d}$ [dividendo]

Proof: Given that,

$\dfrac{a}{b} = \dfrac{c}{d}$

or, $\dfrac{a}{b} - 1 = \dfrac{c}{d} - 1$ [subtracting 1 from both the sides]

i.e., $\dfrac{a-b}{b} = \dfrac{c-d}{d}$

5. If $a : b = c : d$, then $\dfrac{a+b}{a-b} = \dfrac{c+d}{c-d}$ [componendo-Dividendo]

Proof: Given that, $a : b = c : d$

or, $\dfrac{a}{b} = \dfrac{c}{d}$

By componendo, $\dfrac{a+b}{b} = \dfrac{c+d}{d} \ldots (1)$

Again,by dividendo, $\dfrac{a-b}{b} = \dfrac{c-d}{d}$

or, $\dfrac{b}{a-b} = \dfrac{d}{c-d}$ [[by invertendo]] $\ldots (2)$

Therefore, $\dfrac{a+b}{b} \times \dfrac{b}{a-b} = \dfrac{c+d}{d} \times \dfrac{d}{c-d}$ [multiplying(1) & (2)]

i.e., $\dfrac{a+b}{a-b} = \dfrac{c+d}{c-d}$ [here $a \neq b, c \neq d$]

6. If $\dfrac{a}{b} = \dfrac{c}{d} = \dfrac{e}{f} = \dfrac{g}{h}$, then each of the ratio $= \dfrac{a+c+e+g}{b+d+f+h}$

Proof: let,

$\dfrac{a}{b} = \dfrac{c}{d} = \dfrac{e}{f} = \dfrac{g}{h} = k$

$\therefore a = bk,\ c = dk,\ e = fk,\ g = hk$

$$\therefore \frac{a+c+e+g}{b+d+f+h} = \frac{bk+dk+fk+hk}{b+d+f+h} = \frac{k(b+d+f+h)}{b+d+f+h} = k$$

But, k is equal to each ratio in the given proportion.

$$\therefore \frac{a}{b} = \frac{c}{d} = \frac{e}{f} = \frac{g}{h} = \frac{a+c+e+g}{b+d+f+h}$$

Work:

1) The sum of ages of mother and daughter is s years. The ratio of their ages, before t years was $r : p$. After x years, what will be the ratio of their ages?

2) Let a man be standing at p metre distance from a light-post, r be the height of the man, s be the shadow length. Determine the height of the light-post in terms of p, r and s.

Example 2. The ratio of present ages of father and son is $7 : 2$, and after 5 years, the ratio will be $8 : 3$. What are their present ages?

Solution: Let the present age of father be a and that of the son be b.

So, by the conditions of first and second of the problems, we have,

$$\frac{a}{b} = \frac{7}{2} \ldots (1)$$

$$\frac{a+5}{b+5} = \frac{8}{3} \ldots (2)$$

From equation(1),we get,

$$a = \frac{7b}{2} \ldots (3)$$

From equation(2),we get,

$3(a+5) = 8(b+5)$

or, $3a + 15 = 8b + 40$

or, $3a - 8b = 40 - 15$

or, $3 \times \frac{7b}{2} - 8b = 25$ [by using(3)]

or, $\frac{21b - 16b}{2} = 25$

or, $5b = 50$

$\therefore b = 10$

In equation (3), by putting $b = 10$, we get $a = \dfrac{7 \times 10}{2} = 35$.

$\therefore$ The present age of father is 35 years, and that of the son is 10 years.

Example 3. If $a : b = b : c$, prove that, $\left(\dfrac{a+b}{b+c}\right)^2 = \dfrac{a^2+b^2}{b^2+c^2}$

Solution: Given that, $a : b = b : c$

$\therefore b^2 = ac$

Now, $\left(\dfrac{a+b}{b+c}\right)^2 = \dfrac{(a+b)^2}{(b+c)^2}$

$= \dfrac{a^2+2ab+b^2}{b^2+2bc+c^2}$

$= \dfrac{a^2+2ab+ac}{ac+2bc+c^2}$

$= \dfrac{a(a+2b+c)}{c(a+2b+c)}$

$= \dfrac{a}{c}$

And, $\dfrac{a^2+b^2}{b^2+c^2} = \dfrac{a^2+ac}{ac+c^2}$

$= \dfrac{a(a+c)}{c(a+c)}$

$= \dfrac{a}{c}$

$\therefore \left(\dfrac{a+b}{b+c}\right)^2 = \dfrac{a^2+b^2}{b^2+c^2}$

Example 4. If $\dfrac{a}{b} = \dfrac{c}{d}$, show that, $\dfrac{a^2+b^2}{a^2-b^2} = \dfrac{ac+bd}{ac-bd}$

Solution: Let, $\dfrac{a}{b} = \dfrac{c}{d} = k$

$\therefore a = bk$ and $c = dk$

Now, $\dfrac{a^2+b^2}{a^2-b^2} = \dfrac{(bk)^2+b^2}{(bk)^2-b^2} = \dfrac{b^2(k^2+1)}{b^2(k^2-1)} = \dfrac{k^2+1}{k^2-1}$

And, $\dfrac{ac+bd}{ac-bd} = \dfrac{bk \cdot dk + bd}{bk \cdot dk - bd} = \dfrac{bd(k^2+1)}{bd(k^2-1)} = \dfrac{k^2+1}{k^2-1}$

$\therefore \dfrac{a^2+b^2}{a^2-b^2} = \dfrac{ac+bd}{ac-bd}$

Example 5. Solve: $\dfrac{1-ax}{1+ax}\sqrt{\dfrac{1+bx}{1-bx}} = 1$ where $0 < b < 2a < 2b$

Solution: Given that, $\dfrac{1-ax}{1+ax}\sqrt{\dfrac{1+bx}{1-bx}} = 1$

or, $\sqrt{\dfrac{1+bx}{1-bx}} = \dfrac{1+ax}{1-ax}$

or, $\dfrac{1+bx}{1-bx} = \dfrac{(1+ax)^2}{(1-ax)^2}$ [squaring both the sides]

or, $\dfrac{1+bx}{1-bx} = \dfrac{1+2ax+a^2x^2}{1-2ax+a^2x^2}$

or, $\dfrac{1+bx+1-bx}{1+bx-1+bx} = \dfrac{1+2ax+a^2x^2+1-2ax+a^2x^2}{1+2ax+a^2x^2-1+2ax-a^2x^2}$ [by componendo and dividendo]

or, $\dfrac{2}{2bx} = \dfrac{2(1+a^2x^2)}{4ax}$

or, $2ax = bx(1+a^2x^2)$

or, $x\{2a - b(1+a^2x^2)\} = 0$

$\therefore\ x = 0$

Again, $2a - b(1+a^2x^2) = 0$

or, $b(1+a^2x^2) = 2a$

or, $1+a^2x^2 = \dfrac{2a}{b}$

or, $a^2x^2 = \dfrac{2a}{b} - 1$

or, $x^2 = \dfrac{1}{a^2}\left(\dfrac{2a}{b} - 1\right)$

$\therefore x = \pm\dfrac{1}{a}\sqrt{\dfrac{2a}{b} - 1}$

Thus, the required solution is $x = 0, \pm\dfrac{1}{a}\sqrt{\dfrac{2a}{b} - 1}$

2026

Example 6. If $\dfrac{\sqrt{1+x}+\sqrt{1-x}}{\sqrt{1+x}-\sqrt{1-x}} = p$, prove that, $p^2 - \dfrac{2p}{x} + 1 = 0$

Solution: Given that, $\dfrac{\sqrt{1+x}+\sqrt{1-x}}{\sqrt{1+x}-\sqrt{1-x}} = p$

or, $\dfrac{\sqrt{1+x}+\sqrt{1-x}+\sqrt{1+x}-\sqrt{1-x}}{\sqrt{1+x}+\sqrt{1-x}-\sqrt{1+x}+\sqrt{1-x}} = \dfrac{p+1}{p-1}$ [by componendo - dividendo]

or, $\dfrac{2\sqrt{1+x}}{2\sqrt{1-x}} = \dfrac{p+1}{p-1}$

or, $\dfrac{\sqrt{1+x}}{\sqrt{1-x}} = \dfrac{p+1}{p-1}$

or, $\dfrac{1+x}{1-x} = \dfrac{(p+1)^2}{(p-1)^2} = \dfrac{p^2+2p+1}{p^2-2p+1}$ [squaring both the sides]

or, $\dfrac{1+x+1-x}{1+x-1+x} = \dfrac{p^2+2p+1+p^2-2p+1}{p^2+2p+1-p^2+2p-1}$ [by componendo - dividendo]

or, $\dfrac{2}{2x} = \dfrac{2(p^2+1)}{4p}$

or, $\dfrac{1}{x} = \dfrac{p^2+1}{2p}$

or, $p^2+1 = \dfrac{2p}{x}$

$\therefore p^2 - \dfrac{2p}{x} + 1 = 0$

Example 7. If $\dfrac{a^3+b^3}{a-b+c} = a(a+b)$, prove that, a, b, c are continued proportion.

Solution: Given that, $\dfrac{a^3+b^3}{a-b+c} = a(a+b)$

or, $\dfrac{(a+b)(a^2-ab+b^2)}{a-b+c} = a(a+b)$

or, $\dfrac{a^2-ab+b^2}{a-b+c} = a$ [dividing both the sides by $(a+b)$]

or, $a^2-ab+b^2 = a^2-ab+ac$

or, $b^2 = ac$

$\therefore a, b, c$ are continued proportion.

Example 8. If $\dfrac{a+b}{b+c}=\dfrac{c+d}{d+a}$, prove that, $c=a$ or $a+b+c+d=0$

Solution: Given that, $\dfrac{a+b}{b+c}=\dfrac{c+d}{d+a}$

or, $\dfrac{a+b}{b+c}-1=\dfrac{c+d}{d+a}-1$ [subtracting 1 from both the sides]

or, $\dfrac{a+b-b-c}{b+c}=\dfrac{c+d-d-a}{d+a}$

or, $\dfrac{a-c}{b+c}=-\dfrac{a-c}{d+a}$

or, $\dfrac{a-c}{b+c}+\dfrac{a-c}{d+a}=0$

or, $(a-c)\left(\dfrac{1}{b+c}+\dfrac{1}{d+a}\right)=0$

or, $(a-c)\dfrac{(d+a+b+c)}{(b+c)(d+a)}=0$

or, $(a-c)\ (d+a+b+c)=0$

$\therefore\ a-c=0$ or, $d+a+b+c=0$

$\therefore\ c=a$ or, $a+b+c+d=0$

Example 9. If $\dfrac{x}{y+z}=\dfrac{y}{z+x}=\dfrac{z}{x+y}$ and x, y, z are not mutually equal, prove that the value of each ratio is either equal to -1 or equal to $\dfrac{1}{2}$

Solution: Let, $\dfrac{x}{y+z}=\dfrac{y}{z+x}=\dfrac{z}{x+y}=k$

$\therefore x=k(y+z)\ldots(1)$

$y=k(z+x)\ldots(2)$

$z=k(x+y)\ldots(3)$

By subtracting equation (2) from (1),

$x-y=k(y-x)$ or, $k(y-x)=-(y-x)$

$\therefore\ k=-1$

Again, adding equations (1), (2) and (3), we get,

$x+y+z=k(y+z+z+x+x+y)=2k(x+y+z)$

or, $k = \dfrac{(x+y+z)}{2(x+y+z)}$

$\therefore\ k = \dfrac{1}{2}$

$\therefore$ The value of each of the ratio is -1 or $\dfrac{1}{2}$.

Example 10. If $ax = by = cz$, show that, $\dfrac{x^2}{yz} + \dfrac{y^2}{zx} + \dfrac{z^2}{xy} = \dfrac{bc}{a^2} + \dfrac{ca}{b^2} + \dfrac{ab}{c^2}$

Solution: Let, $ax = by = cz = k$

$\therefore x = \dfrac{k}{a},\ y = \dfrac{k}{b},\ z = \dfrac{k}{c}$

Now, $\dfrac{x^2}{yz} + \dfrac{y^2}{zx} + \dfrac{z^2}{xy} = \dfrac{k^2}{a^2} \times \dfrac{bc}{k^2} + \dfrac{k^2}{b^2} \times \dfrac{ca}{k^2} + \dfrac{k^2}{c^2} \times \dfrac{ab}{k^2} = \dfrac{bc}{a^2} + \dfrac{ca}{b^2} + \dfrac{ab}{c^2}$

That is, $\dfrac{x^2}{yz} + \dfrac{y^2}{zx} + \dfrac{z^2}{xy} = \dfrac{bc}{a^2} + \dfrac{ca}{b^2} + \dfrac{ab}{c^2}$

Example 11. If a, b, c and d are continued proportional, and $x = \dfrac{10pq}{p+q}$

1) Show that, $\dfrac{a}{c} = \dfrac{a^2+b^2}{b^2+c^2}$
2) Prove that, $(a^2+b^2+c^2)(b^2+c^2+d^2) = (ab+bc+cd)^2$
3) Determine $\dfrac{x+5p}{x-5p} + \dfrac{x+5q}{x-5q}$,where $p \neq q$

Solution:

1) Given that, $a : b = b : c$ or, $\dfrac{a}{b} = \dfrac{b}{c}$ or, $ac = b^2$

   RHS $= \dfrac{a^2+b^2}{b^2+c^2} = \dfrac{a^2+ac}{ac+c^2} = \dfrac{a(a+c)}{c(a+c)} = \dfrac{a}{c} =$ LHS

   $\therefore\ \dfrac{a}{c} = \dfrac{a^2+b^2}{b^2+c^2}$

2) Given that, a, b, c and d are continued proportional.

   $\therefore \dfrac{a}{b} = \dfrac{b}{c} = \dfrac{c}{d}$

   Let, $\dfrac{a}{b} = \dfrac{b}{c} = \dfrac{c}{d} = k$, where k is a constant proportional.

   $\therefore\ \dfrac{c}{d} = k$ or, $c = dk$

   $\dfrac{b}{c} = k$ or, $b = ck = dk \cdot k = dk^2$

$\frac{a}{b} = k$ or, $a = bk = dk^2 \cdot k = dk^3$

LHS $= (a^2 + b^2 + c^2)(b^2 + c^2 + d^2)$

$= \{(dk^3)^2 + (dk^2)^2 + (dk)^2\}\{(dk^2)^2 + (dk)^2 + d^2\}$

$= (d^2k^6 + d^2k^4 + d^2k^2)(d^2k^4 + d^2k^2 + d^2)$

$= d^2k^2(k^4 + k^2 + 1)d^2(k^4 + k^2 + 1)$

$= d^4k^2(k^4 + k^2 + 1)^2$

RHS $= (ab + bc + cd)^2$

$= (dk^3 \cdot dk^2 + dk^2 \cdot dk + dk \cdot d)^2$

$= (d^2k^5 + d^2k^3 + d^2k)^2$

$= \{d^2k(k^4 + k^2 + 1)\}^2$

$= d^4k^2(k^4 + k^2 + 1)^2 =$ LHS

$\therefore\ (a^2 + b^2 + c^2)(b^2 + c^2 + d^2) = (ab + bc + cd)^2$

3) Given that, $x = \frac{10pq}{p+q}$

or, $\frac{x}{5p} = \frac{2q}{p+q}$

or, $\frac{x+5p}{x-5p} = \frac{2q+p+q}{2q-p-q}$ [by componendo-dividendo]

or, $\frac{x+5p}{x-5p} = \frac{p+3q}{q-p}$ ... (1)

Again, $x = \frac{10pq}{p+q}$

or, $\frac{x}{5q} = \frac{2p}{p+q}$

or, $\frac{x+5q}{x-5q} = \frac{2p+p+q}{2p-p-q}$ [by componendo-dividendo]

or, $\frac{x+5q}{x-5q} = \frac{3p+q}{p-q}$ ... (2)

Here, by adding equations (1) and (2) we get,

$\frac{x+5p}{x-5p} + \frac{x+5q}{x-5q} = \frac{p+3q}{q-p} + \frac{3p+q}{p-q} = \frac{p+3q}{q-p} - \frac{3p+q}{q-p}$

$= \frac{p+3q-3p-q}{q-p} = \frac{2q-2p}{q-p} = \frac{2(q-p)}{q-p} = 2 \quad [\because\ q-p \neq 0]$

2026

Exercise 11.1

1. If the sides of two squares are a and b metre respectively, what will be the ratio of their areas?
2. If the area of a circle is equal to the area of a square, find the ratio of their perimeter.
3. If the ratio of two numbers is $3 : 4$ and their L.C.M. is 180, find out the two numbers.
4. The ratio of absent and present students of a day in your class is found to be $1 : 4$. Express the number of absent students in percentage with respect to the total number of students.
5. A thing is first bought and then sold with 28% loss. Find the ratio of buying and selling cost.
6. Sum of the ages of father and son is 70 years. 7 years ago the ratio of their ages were $5 : 2$. What will the ratio of their ages after 5 years?
7. If $a : b = b : c$, prove that,

   1) $\dfrac{a}{c} = \dfrac{a^2 + b^2}{b^2 + c^2}$

   2) $a^2b^2c^2\left(\dfrac{1}{a^3} + \dfrac{1}{b^3} + \dfrac{1}{c^3}\right) = a^3 + b^3 + c^3$

   3) $\dfrac{abc(a+b+c)^3}{(ab+bc+ca)^3} = 1$

8. Solve:

   1) $\dfrac{1-\sqrt{1-x}}{1+\sqrt{1-x}} = \dfrac{1}{3}$

   2) $\dfrac{a+x-\sqrt{a^2-x^2}}{a+x+\sqrt{a^2-x^2}} = \dfrac{b}{x}$, $2a > b > 0$ and $x \neq 0$

   3) $81\left(\dfrac{1-x}{1+x}\right)^3 = \dfrac{1+x}{1-x}$

9. if $\dfrac{a}{b} = \dfrac{b}{c} = \dfrac{c}{d}$, show that,

   1) $\dfrac{a^3+b^3}{b^3+c^3} = \dfrac{b^3+c^3}{c^3+d^3}$

   2) $(a^2+b^2+c^2)(b^2+c^2+d^2) = (ab+bc+cd)^2$

10. If $x = \dfrac{4ab}{a+b}$, show that, $\dfrac{x+2a}{x-2a} + \dfrac{x+2b}{x-2b} = 2,\ a \neq b$

11. If $x = \dfrac{\sqrt[3]{m+1}+\sqrt[3]{m-1}}{\sqrt[3]{m+1}-\sqrt[3]{m-1}}$, prove that, $x^3 - 3mx^2 + 3x - m = 0$

12. If $x = \dfrac{\sqrt{2a+3b}+\sqrt{2a-3b}}{\sqrt{2a+3b}-\sqrt{2a-3b}}$, show that, $3bx^2 - 4ax + 3b = 0$

13. If $\dfrac{a^2+b^2}{b^2+c^2} = \dfrac{(a+b)^2}{(b+c)^2}$,show that, a, b, c are continued proportion.

14. If $\dfrac{x}{b+c} = \dfrac{y}{c+a} = \dfrac{z}{a+b}$, prove that, $\dfrac{a}{y+z-x} = \dfrac{b}{z+x-y} = \dfrac{c}{x+y-z}$.

15. If $\dfrac{bz-cy}{a} = \dfrac{cx-az}{b} = \dfrac{ay-bx}{c}$, prove that, $\dfrac{x}{a} = \dfrac{y}{b} = \dfrac{z}{c}$.

16. If $\dfrac{a+b-c}{a+b} = \dfrac{b+c-a}{b+c} = \dfrac{c+a-b}{c+a}$ and $a+b+c \neq 0$, prove that, $a = b = c$

17. If $\dfrac{x}{xa+yb+zc} = \dfrac{y}{ya+zb+xc} = \dfrac{z}{za+xb+yc}$ and $x+y+z \neq 0$, show that each of the ratio is $= \dfrac{1}{a+b+c}$.

18. If $(a+b+c)p = (b+c-a)q = (c+a-b)r = (a+b-c)s$, then prove that, $\dfrac{1}{q} + \dfrac{1}{r} + \dfrac{1}{s} = \dfrac{1}{p}$.

19. If $lx = my = nz$, then show that $\dfrac{x^2}{yz} + \dfrac{y^2}{zx} + \dfrac{z^2}{xy} = \dfrac{mn}{l^2} + \dfrac{nl}{m^2} + \dfrac{lm}{n^2}$.

20. If $\dfrac{p}{q} = \dfrac{a^2}{b^2}$ and $\dfrac{a}{b} = \dfrac{\sqrt{a+q}}{\sqrt{a-q}}$,show that, $\dfrac{p+q}{a} = \dfrac{p-q}{q}$.

Continued Ratio

Let us assume that Roni's earning is Tk 1000, Soni's earning is Tk 1500 and Samir's earning is Tk 2500. Here, Roni's earning : Soni's earning = 1000 : 1500 =

2 : 3; Soni's earning : Samir's earning = 1500 : 2500 = 3 : 5. Hence, Roni's earning : Soni's earning : Sami's earning = 2 : 3 : 5.

If two ratios are of the form $a : b$ and $b : c$, they can be re-written in the form $a : b : c$. This is called continued ratio. Any two or more ratios can be expressed in this form. It is noted that if two ratios are to be expressed in the form $a : b : c$, the antecedent of the first ratio and the subsequent of the second ratio need to be equal. For example, if two ratios 2 : 3 and 4 : 3 are to be put in the form $a : b : c$, the subsequent quantity of the first ratio is to be made equal to the antecedent quantity of the second ratio. That is, those quantities are to be made equal to their L.C.M.

Here, L.C.M of 3 and 4 is 12.

Here, $2 : 3 = \frac{2}{3} = \frac{2 \times 4}{3 \times 4} = \frac{8}{12} = 8 : 12$

Again, $4 : 3 = \frac{4}{3} = \frac{4 \times 3}{3 \times 3} = \frac{12}{9} = 12 : 9$

Therefore, if the ratios 2 : 3 and 4 : 3 are expressed in the form, $a : b : c$ will be 8 : 12 : 9

It is to be noted that if the earning of Samir in the above example is 1125, the ratio of their earnings will be 8 : 12 : 9.

Example 12. If a, b, c are quantities of same kind and $a : b = 3 : 4$, $b : c = 6 : 7$, what will be $a : b : c$?

Solution: $a : b = \frac{3}{4} = \frac{3 \times 3}{4 \times 3} = \frac{9}{12}$ and $b : c = \frac{6}{7} = \frac{6 \times 2}{7 \times 2} = \frac{12}{14}$ [LC.M. of 4 and 6 is 12]]

$\therefore a : b : c = 9 : 12 : 14$

Example 13. The ratio of angles of a triangle is 3 : 4 : 5. Express the angles in degree.

Solution: Let the angles, according to given ratio, be $3x$, $4x$ and $5x$. Sum of three angles = 108°. According to the problem, $3x + 4x + 5x = 180°$ or, $12x = 180°$ or, $x = 15°$, Therefore, the angles are $3x = 3 \times 15° = 45°$

$4x = 4 \times 15° = 60°$

and $5x = 5 \times 15° = 75°$

Example 14. If the sides of a square increase by 10%, how much will the area be increased in percentage?

Solution: Let, each side of the square be a metre. Therefore, the area of the square be a^2 square metre. If the side of the square increases by 10%, each side will be ($a + a$ of 10%) metre, or $1.10a$ metre. In this case, the area of the square will be $(1.10a)^2$ square metre, or $1.21a^2$ square metre. Thus the the area increases by $(1.21a^2 - a^2) = 0.21a^2$ square metre.

$\therefore$ The percentage of increment of the area will be $\dfrac{0.21a^2}{a^2} \times 100\% = 21\%$

Work:

1) There are 35 male and 25 female students in your class. The ratio of rice and pulse given by each of the male and female students for making Khicuri in a picnic is 3 : 1 and 5 : 2, respectively. Find the ratio of total rice and total pulse.
2) The amount of pulse, mustard and paddy produced in a farmer's farm are 75 kg, 100 kg and 525 kg respectively. The grains are sold at price of 100, 120 and 30 taka respectively. After selling all the grains, calculate the ratio of income from the individual grain.

Proportional Division

Division of a quantity into fixed ratio is called proportional division. If S is to be divided in a given ratio $a : b : c : d$, dividing S by $a+b+c+d$, the parts a, b, c and d need to be taken. Therefore, 1st part $= S$ of $\dfrac{a}{a+b+c+d} = \dfrac{Sa}{a+b+c+d}$

2nd part $= S$ of $\dfrac{b}{a+b+c+d} = \dfrac{Sb}{a+b+c+d}$

3rd part $= S$ of $\dfrac{c}{a+b+c+d} = \dfrac{Sc}{a+b+c+d}$

4th part $= S$ of $\dfrac{d}{a+b+c+d} = \dfrac{Sd}{a+b+c+d}$

In this way, any quantity may be divided into any fixed ratio.

Example 15. The area of a rectangular land is 12 hectares and length of its diagonal is 500 metres. The ratio of length and breadth of this land to that of other land are 3 : 4 and 2 : 3 respectively.

1) What is the area of the given land in square metre?
2) What is the area of the other land?
3) What is the breadth of the given land?

Solution:

1) We know, 1 hectare $= 10,000$ square metres.

   $\therefore$ 12 hectares $= 12 \times 10,000 = 120000$ square metres

2) Given that, the ratio of length and breadth of the given land to that of other land are 3 : 4 and 2 : 3 respectively.

   Let length of the given land be $3x$ metres and breadth be $2y$ metres.

   Therefore, the length of the other land is $4x$ metres and breadth is $3y$ metres.

   $\therefore$ the area of the given land is $= 3x \cdot 2y = 6xy$ square metres and the area of the other land is $= 4x \cdot 3y = 12xy$ square metres.

   According to the given question, $6xy = 120000$ or, $xy = 20000$ $\therefore$ the area of the other land is $= 12xy = 12 \times 20000 = 240000$ square metres.

3) Let the length of the given land be $3x$ metres and the breadth be $2y$ metres.

   Therefore, the length of its one diagonal is $\sqrt{(3x)^2 + (2y)^2}$ metres.

   We get from the previous (b) problem, $xy = 20000$

   According to the question, $\sqrt{(3x)^2 + (2y)^2} = 500$

   or, $9x^2 + 4y^2 = 250000$

   or, $(3x + 2y)^2 - 2 \cdot 3x \cdot 2y = 250000$

   or, $(3x + 2y)^2 - 12xy = 250000$

   or, $(3x + 2y)^2 - 12 \times 20000 = 250000$

   or, $(3x + 2y)^2 = 250000 + 240000$

   or, $(3x + 2y)^2 = 490000$

   or, $3x + 2y = 700 \cdots (1)$

Again, $(3x-2y)^2=(3x+2y)^2-4\cdot 3x\cdot 2y$

or, $(3x-2y)^2=(3x+2y)^2-24xy$

or, $(3x-2y)^2=(700)^2-24\times 20000$

or, $(3x-2y)^2=490000-480000$

or, $(3x-2y)^2=10000$

or, $3x-2y=100 \ldots (2)$

By subtracting (2) from (1) we get,

$4y=600$ or, $y=150$

$\therefore$ the breadth of the given land is $2y=2\times 150=300$ metres.

Exercise 11.2

1. If a, b, c are continued proportional, which one of the followings is correct?
   1) $a^2 = bc$ 2) $b^2 = ac$
   3) $ab = bc$ 4) $a = b = c$

2. The ratio of ages of Arif and Akib is 5 : 3. If Arif is of 20 years old, how many years later the ratio of their ages will be 7 : 5?
   1) 5 years 2) 6 years
   3) 8 years 4) 10 years

3. If $x : y = 7 : 5$, $y : z = 5 : 7$, then $x : z =$ how much ?
   1) 35 : 49 2) 35 : 35
   3) 25 : 49 4) 49 : 25

4. If $x : y = 2 : 1$ and $y : z = 2 : 1$,then–
   (i) x, y, z are continued proportional
   (ii) $z : x = 1 : 4$
   (iii) $y^2 + zx = 4yz$
   Which one of the following is correct?
   1) i and ii 2) i and iii 3) ii and iii 4) i, ii and iii

5. If $\dfrac{a}{x} = \dfrac{m^2+n^2}{2mn}$, then $\dfrac{\sqrt{a+x}}{\sqrt{a-x}} =$ what ?
   1) $\dfrac{m}{n}$ 2) $\dfrac{m+n}{m-n}$ 3) $\dfrac{m-n}{m+n}$ 4) $\dfrac{n}{m}$

6. Weight of 1 cubic c.m wood is 7 decigram. What is the percentage of weight of the wood to the weight of equal volume of water?

7. Distribute Tk. 300 among a, b, c, d in such a way that the ratios are a's part : b's part = 2 : 3, b's part : c's part = 1 : 2 and c's part : d's part 3 : 2.

8. Three fishermen have caught 690 pieces of fish. The ratios of their parts are $\frac{2}{3}, \frac{4}{5}$ and $\frac{5}{6}$. How many fish will each of them get?

9. The perimeter of a triangle is 45 cm. The ratio of the lengths of the sides is 3 : 5 : 7. Find the length of each sides.

10. In a cricket game, the total runs scored by Sakib, Mushfique and Mashrafi were 171. The ratio of runs scored by Sakib and Mushfique, and Mushfique and Mashrafi was 3 : 2. What were the runs scored by them individually?

11. In an office, there were 2 officers, 7 clarks and 3 bearers. If a bearer gets Tk. 1, a clerk gets Tk. 2 and an officer gets Tk. 4. Their total salary is Tk. $150,000$. What is their individual salary?

12. If the sides of a square are increased by 20%, what is percentage of increment of the area of the square?

13. If the length of a rectangle is increased by 10% and the breadth is decreased by 10%, what is the percentage of increase or decrease of the area of the rectangle?

14. In a field, the ratio of production is 4 : 7 before and after irrigation. In that field, the production of paddy in a land previously was 304 quintal. What would be the production of paddy after irrigation?

15. If the ratio of paddy and rice produced from paddy is 3 : 2 and the ratio of wheat and suzi produced from wheat is 4 : 3, find the ratio of rice and suzi produced from equal quantity of rice and wheat.

16. The area of a land is 432 square metre. If the ratios of lengths and breadths of that land and that of another land are 3 : 4 and 2 : 5 respectively, then what is the area of another land?

17. Zami and Simi take loans of different amounts at the rate of 10% simple profit on the same day from the same Bank. The amount on capital and profit which Zami refunds after 2 years is the the same amount that Simi refunds after 3 years on capital and profit. Find the ratio of their loan.

18. A rectangular solid has length, width, and height x, y & z $(x > y > z)$ and $\frac{(y^2+z^2)}{(y+z)^2} = \frac{(x^2+y^2)}{(x+y)^2}$ respectively. The height of the solid is reduced by 10%.

    a) Find the ratio of selling price to cost price if a product is sold at a 10% profit.

    b) Show that $y^2 = zx$

    c) Find the percentage by which the volume of the solid decreases.

2026

Sample Questions

Multiple Choice Questions:

1. If the sides of a square double, how much will the area of a square be increased?

   (a) 2 times (b) 3 times (c) 4 times (d) 6 times

2. If b, a, c are continued proportional, then−

   (i) $a^2 = bc$ (ii) $\frac{b}{a} = \frac{c}{a}$ (iii) $\frac{a+b}{a-b} = \frac{c+a}{c-a}$

   Which one of the following is correct?

   (a) i & ii (b) i & iii (c) ii & iii (d) i, ii & iii

If the sides of a triangle are in the ratio $3 : 4 : 5$ and the perimeter is 36 cm, then answer the following Questions 3 & 4:

3. What is the difference between the largest and smallest sides of the triangle?

   (a) 9 (b) 6 (c) 5 (d) 2

4. What is the area of the triangle in square centimeters?

   (a) 6 (b) 54 (c) 67 (d) 90

Creative Question:

5. **Scenario 1:** One day in a class, the ratio of absent to present students was 1:4. If 5 more students had been present, the ratio would have been 1:9.
   Scenario 2: An area of a rectangular field is 336 square units. The ratio of the length and width of another rectangular field to this one is 2:3 and 4:5 respectively.

   a) A product is bought for Tk. 400 and sold at a 10% loss. Find the ratio of selling price to cost price.
   b) Find the total number of students in the class in Scenario 1.
   c) Find the area of the second field in Scenario 2.

Short-Answer Questions:

6. a) If the ratio of two numbers is 5:7 and their H.C.F. is 4, what is L.C.M. of the numbers?
   b) If $\frac{x^2+y^2}{y^2+z^2} = \frac{x}{z}$ and $z \neq x$ show that, x, y and z are in continued proportion.
   c) If the mass of 1 cubic centimeter of wood is 700 mg, find the ratio of mass of the same volume of wood to that of water.
   d) If $x : y = 3 : 4$ and $y : z = 2 : 3$ show that $\frac{x}{3} = \frac{y}{4} = \frac{z}{6}$.